Topological Sorting of Hasse Diagram (DAGs / Directed Acyclic Graphs)

Q: Partial order $\Longrightarrow$ Total order.

Catch?

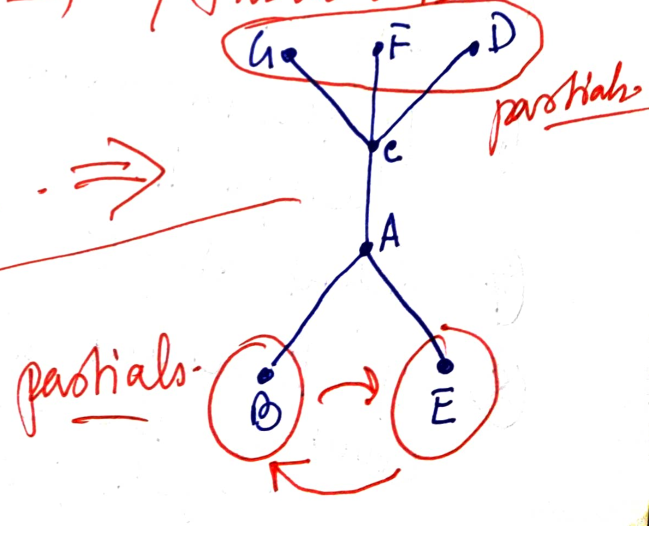
Claims.

1 partial order $\Rightarrow$ 1 total order ✗

1 partial order gives rise to multiple total order $\longrightarrow$ getting one of them is topological sorting.

E.g. Assembling a toy $\rightarrow$ 7 tasks to be solved.

Tasks: A, B, C, . . . G . $\Longrightarrow$



extract a total order.

in-degree → use this to develop a total order from a partial order.

Kahn's Algorithm.

1. Find in-degree of all vertices and store it in ~~a sequence~~. an array
2. Pick all vertices having zero in-degree and enque them in a queue.
3. Dequeue a vertex from the front of the queue and place it in the result array
4. Increase the count of visited nodes by 1.
5. Reduce the in-degree of neighbouring vertices by 1.
6. If the in-degree of any vertex becomes zero then enque it.
7. Repeat step 3
8. If the count of visited nodes is NOT equal to the count of the vertices in the graph then there is a cycle; ordering is not possible.

Maximal & Minimal elements.

If $(A, \preceq)$ is a poset then an element $x \in A$ is called a maximal element of A if for all $a \in A$ $a \neq x \Rightarrow x \npreceq a$

No element $a \in A$ that covers x.

An element $y \in A$ is called a minimal element of A if whenever $b \in A$ and $b \neq y$ then $b \npreceq y$.

No one else in A is covered by the minimal element y.

E.g. $U = \{1, 2, 3\}$, $A = P(U)$.

$\preceq \rightarrow \subseteq$

Minimal element: Φ

Maximal element: U

Theorem:

If $(A, \leq)$ is a poset & A is finite then A has both maximal & minimal element.

Minimal:

① A is finite.

② Pick any $a_1 \in A$. If a_1 is minimal stop.

③ If a_1 is not minimal there will be some $a_2 \in A$ s.t. $a_2 \leq a_1$. If a_2 is minimal, stop. Otherwise there must be $a_3 \in A$, s.t. $a_3 \leq a_2$

. . . . should all be

$a_1 \neq a_2 \neq a_3 \ldots$

distinct.

If not →

cycle: $a_i \ldots a_i$ | all elements in this chain must be equal by the antisymmetry property.

④ A is finite → we can do the above at most $|A|$ times.

⑤ A some point there will be no element below an element a_k (minimal) $\square$.

Greatest & least elements.

If $(A, \leq)$ then an element $x \in A$ is called a least element if $x \leq a$ for $\forall a \in A$.

If $(A, \leq)$ then an element $y \in A$ is called the greatest element $a \leq y$ for $\forall a \in A$.

Theorem: P.T. the greatest & least element in a poset must be unique.

By way contradiction:

Let these two greatest elements y & y'

$\therefore \boxed{y \leq y' \quad \& \quad y' \leq y}$

by antisymmetric property

$y = y'$

Upper bounds & Lower bounds.

$(A, \leq)$ & $B \subseteq A$.

An element $x \in A$ is called a lower bound of B if $x \leq b$, $\forall b \in B$.

An element $y \in A$ is called an upper bound of B if $b \leq y$, $\forall b \in B$.

An element $x' \in A$ is called the greatest lower bound (glb) of B if it is a lower bound of B and if for all other lower bounds $x'' \in B$ we have $x'' \leq x'$

An element $y' \in A$ is called the least upper bound (lub) of B if it is an upper bound of B and if $y' \leq y''$ for all other $y'' \in B$.

entails a ~~sense~~ sense of tightness.

E.g. $\mathcal{U} = \{1,2,3,4\}$, $A = P(\mathcal{U})$, $B = \{\{1\}, \{1,2\}, \{2\}\}$.

$\leq \to \subseteq$.

$lb = $

$glb = \Phi$

$ub = \{\{1,2\}, \{1,2,4\}, \{1,2,3\}, \{1,2,3,4\} \ldots\}$.

$lub = \{1,2\}$.

H.W. The poset $(A, \leq)$ has a subset $B \subseteq A$ then B can have at most one (a) glb (b) lub.

Defn

A poset $(A, \leq)$ is called a lattice if for any $(x,y) \in A$ the element $lub\{x,y\}$, $glb\{x,y\}$ both exist in A.

$A = N$, $x, y \in N$. $\Rightarrow$ $\leq \to \leq$

$lub\{x,y\} = \max\{x,y\}$

$glb\{x,y\} = \min\{x,y\}$.